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Deal or No Deal: The Effects of Financial-Term Disclosure on Investor Evaluations and the Mediating Role of Ambiguity Aversion

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Running Head: Deal or No Deal

Abstract

Entrepreneurs receive conflicting advice about whether to include deal terms—such as the desired valuation, the amount of capital to be raised, and equity to be sold—in startup pitch decks. Some investors argue that transparency regarding desired terms increases credibility, while others caution that it dampens interest. Whether such disclosure helps or harms founders is a theoretically-relevant empirical question, which we address through two randomized controlled trials. Study 1 examines the causal effect of adding a "deal-terms" slide to otherwise identical startup pitch decks. Across 1,008 real investor–pitch evaluations, including deal terms significantly decreases investor favorability across four measures of investment intent, irrespective of whether the terms are favorable or unfavorable. Having established this behavioral effect, in Study 2, we investigate the underlying mechanism by testing seven candidate explanations drawn from prior behavioral decision research: ambiguity aversion, psychological reactance, signaling, cognitive load, decision avoidance, anchoring and adjustment, and loss aversion. In a multilevel experiment manipulating high versus low values for valuation, capital, and equity, we find that only perceived ambiguity acts as a significant mediator: quantitative specificity reduces informational ambiguity, which in turn shapes investor judgments. Collectively, these findings provide evidence on both *whether* and *why* financial term disclosures influence investor judgment, demonstrating that deal-term transparency alters perceived informational clarity rather than the substantive attractiveness of the opportunity.

Keywords: ambiguity aversion; cognitive biases; deal terms; entrepreneurial pitches; randomized controlled trial.

Deal or No Deal: The Effects of Financial-Term Disclosure on Investor Evaluations and the Mediating Role of Ambiguity Aversion

Introduction

Venture creation is a key driver of economic growth and wealth generation, yet its success often depends on securing funding from external investors, especially in the early stages of development (Florida & Kenney, 1988; Roure et al., 1986). At the pre-seed and seed stages, founders rely heavily on pitch decks to convey their ventures to potential investors. These decks closely follow the initial evaluation criteria used by venture capitalists—an early-stage hurdle that entrepreneurs must navigate effectively to launch their ventures (Bernstein et al., 2017; Brooks et al., 2014; Huang & Pearce, 2015). Pitch decks typically outline the venture's team, market opportunity, business model, competitive position, and financial projections (Kalvapalle et al., 2024). Despite their importance, the impact of specific pitch-deck elements on investor interest and evaluation remains largely underexplored and unresolved in the entrepreneurship literature (Gompers et al., 2020; Miloud et al., 2012).

Despite a rapidly growing body of literature on entrepreneurial pitching, most research remains fragmented across separate "unit theories" that focus either on what entrepreneurs do in pitches or on how investors respond, with limited integration between these perspectives (Kalvapalle, Phillips, & Cornelissen, 2024, for a recent review). Existing studies have mainly focused on observable differences in outcomes across gender or race (e.g., Kanze et al., 2018; Younkin & Kuppuswamy, 2017), with comparatively less attention to the underlying psychological mechanisms that drive investor evaluations. Moreover, the communicative dynamics of pitching—how specific disclosure choices shape information processing and bias formation—remain largely under-theorized. By examining the influence of quantitative

investment terms on investor cognition, this study addresses this gap, shedding light on the mechanisms through which pitch content shapes investor judgment, beyond simply documenting outcome disparities.

Specifically, one of the most contentious elements of a startup pitch deck is whether founders should include quantitative deal terms, such as the desired valuation, capital sought, and equity offered. The academic literature offers little guidance on this issue, and entrepreneurs receive conflicting advice from practitioners. Some investors insist that omitting quantitative terms makes a pitch incomplete. As *Pitching Angels* puts it: "*You have to give us the key deal terms, the most important of which, by far, is the valuation (or cap). It's essentially the price you're selling the stock at. It's impossible to evaluate the investment opportunity without it, so please, please, please include it in the deck*" (Pitching Angels, 2021). Others take the opposite stance. For example, Peter Szymanski, Silicon Valley General Counsel, cautions founders that: "*The pitch deck is to present your business, not offer the deal terms*" (Szymanski, 2021). Accelerators echo this split: While some mentors encourage founders to specify deal parameters to show professionalism (Techstars, 2022), others warn that doing so gives investors a reason to pass too soon (Y Combinator, 2020). Thus, whether including quantitative deal terms helps or harms founders remains an important theoretical puzzle as well as an empirical question of practical relevance.

We address this question through two complementary randomized controlled trials. Study 1 tests the causal effect of including a single "deal-terms" slide in otherwise identical startup pitch decks, sourced from real companies that participated in Techstars, a prominent global startup accelerator (Feld & Cohen, 2019). Across 1,008 investor–pitch evaluations, adding the deal-terms slide significantly reduces investor favorability across four measures of investment intent:

perceived likelihood of success, perceived business quality, willingness to take next steps, and willingness to share the opportunity. The effect is robust across startups, industries, and both favorable and unfavorable deal terms.

Having established this behavioral effect, Study 2 examines *why* it occurs. Drawing on behavioral decision theory, which shows that cognitive biases systematically influence strategic evaluations (Obloj & Zenger, 2017), we test seven candidate mechanisms: ambiguity aversion (Ellsberg, 1961), psychological reactance (Brehm, 1966), signaling theory (Spence, 1973), cognitive load theory (Sweller, 1988), decision avoidance theory (Tversky & Shafir, 1992), the anchoring-and-adjustment heuristic (Tversky & Kahneman, 1974), and loss aversion (Kahneman & Tversky, 1979). Each mechanism presents a different perspective on why the provision of deal terms might influence investor judgments: by reducing uncertainty, provoking reactance, signaling credibility, increasing cognitive strain, deterring choice, biasing reference points, or heightening sensitivity to potential loss.

Study 2 uses a multilevel experiment with 2,592 pitch-level observations across 12 Techstars startup pitches. Here, we focus on manipulating the *magnitude* of the deal terms—i.e., high versus low desired valuation, capital to be raised, and equity to be sold—and assess the seven theorized mechanisms. Intriguingly, the results clearly point to a single dominant pathway: Perceived ambiguity is the only construct that fully mediates the relationship between deal-term framing and investor judgment. That is, including quantitative specificity reduces informational ambiguity, which in turn shapes investor judgments. No other tested mechanisms account for the effect.

Together, these studies provide evidence on *whether* and *why* financial-term disclosure influences investor judgments. Including quantitative deal terms alters investor evaluations by

changing perceptions of informational clarity, rather than through mechanisms such as negotiation resistance, cognitive strain, or loss aversion. Theoretically, this work contributes to the literature on behavioral decision-making in entrepreneurial finance by pinpointing ambiguity aversion as the psychological mechanism linking financial disclosure to investment behaviors (Drover et al., 2018; Huang & Pearce, 2015; Zacharakis & Shepherd, 2001). Moreover, it reinforces the scientific value of randomized field-style experiments in advancing our understanding of investor cognition (Agarwal et al., 2025; Camuffo et al., 2020; Hallen & Piezunka, 2023). For practitioners, our findings suggest that transparency about valuation, raise size, and equity is likely to shape investor interest less by changing the deal itself than by changing how clearly the opportunity is perceived.

Theoretical Development

Investors evaluating early-stage ventures face great uncertainty and information asymmetry (Packard, Clark, & Klein, 2017; Shane & Stuart, 2002; Wiltbank et al., 2009). With few objective performance metrics available, they rely heavily on what founders choose to disclose in pitch decks—the primary medium for screening startup opportunities (Bernstein et al., 2017; Brooks et al., 2014; Huang & Pearce, 2015). Behavioral decision theory has long suggested that when information is ambiguous or incomplete, people employ heuristics and rely on psychological cues to make judgments under uncertainty (Tversky & Kahneman, 1974; Slovic, Fischhoff, & Lichtenstein, 1977). Consistent with evidence that subtle contextual cues and changes in choice architecture systematically shape human judgment (e.g., Dai, Milkman, & Riis, 2014), investors may rely on cognitive heuristics when evaluating early-stage ventures. Yet little is currently known about the mechanisms underlying these processes in the context of entrepreneurial financial decision-making.

Recent integrative reviews further emphasize the urgency of mechanism-based theorizing in pitching research. Kalvapalle et al. (2024) argue that while prior studies have advanced "unit theories" explaining discrete cues such as passion, authenticity, or gendered framing, they rarely focus on or explain the underlying psychological pathways through which such cues translate into investor judgments. As we shall see, our approach responds directly to this call by identifying ambiguity aversion as the key, theoretically grounded process that links information disclosure to investor evaluation.

Within this context, a persistent dispute in entrepreneurship focuses on whether founders should include quantitative deal terms in their decks—the desired valuation, capital to be raised, and equity to be sold. Practitioner guidance varies widely: Some accelerators encourage founders to specify terms to demonstrate preparedness (Feld & Mendelson, 2016; Techstars, 2022), whereas others warn that doing so can prematurely disqualify opportunities or anchor investor perceptions (Y Combinator, 2020; Startup School, 2021). From a rational perspective, including such information should improve decision quality by clarifying expectations and facilitating efficient matching between founders and investors (Spence, 1973; Connelly et al., 2011; Petty & Gruber, 2011). From a behavioral perspective, however, presenting explicit numbers could shift investors' psychology in ways that inadvertently reduce interest—for example, by constraining negotiation flexibility, increasing perceived risk, or distracting from qualitative cues (Tversky & Kahneman, 1974; Kahneman & Tversky, 1979; Lichtenstein & Slovic, 2006). This conflicting practitioner advice reflects a deeper theoretical uncertainty: *What psychological processes drive investors' reactions to quantitative deal disclosures?*

In this work, we extend the current literature by experimentally isolating the effect of a common, yet theoretically overlooked, element of entrepreneurial communication—the inclusion

of quantitative deal terms—on investor evaluations, and by identifying the psychological mechanism underlying this effect. Because the current literature offers multiple, parceled, and non-exclusive explanations, we undertook a two-stage experimental approach. Study 1 begins empirically, testing whether the inclusion of deal terms changes investor favorability, without presupposing any mechanism. Finding a reliable causal effect would indicate that this design choice in startup communication meaningfully affects investor judgment. Having established the effect, Study 2 turns to theory development, testing alternative candidate explanations derived from classic behavioral decision research: ambiguity aversion (Ellsberg, 1961; Camerer & Weber, 1992), psychological reactance (Brehm, 1966), signaling (Spence, 1973), cognitive load (Sweller, 1988), decision avoidance (Tversky & Shafir, 1992), anchoring (Tversky & Kahneman, 1974), and loss aversion (Kahneman & Tversky, 1979).

Study 1

Conceptual Framework and Hypotheses

Pitch decks can shape investors' views of a startup's opportunity, quality, and readiness for funding (Bernstein et al., 2017; Brooks et al., 2014; Huang & Pearce, 2015). Yet founders face a strategic choice about whether to include quantitative deal terms—the desired valuation, the amount of capital to be raised, and the proportion of equity to be sold. Theoretically, from a rational-information / signaling perspective, presenting explicit quantitative terms could clarify expectations, facilitate price discovery, and signal professionalism and transaction readiness, which should all increase investor favorability (Petty and Gruber, 2021). Yet, from a behavioral-decision viewpoint, adding numerical values could narrow interpretive flexibility, shift attention from venture quality to negotiation outcomes, and heighten evaluative scrutiny, which should decrease investor favorability (Tversky & Kahneman, 1974; Lichtenstein & Slovic, 2006).

Practitioner advice reflects this split, turning the direction of the effect into a highly relevant empirical question. Therefore, these arguments lead to two alternative hypotheses:

H1a (Rational/Signaling prediction): Including a "deal-terms" slide (i.e., desired valuation, capital to be raised, equity to be sold) in a pitch deck will increase investor favorability toward a startup.

H1b (Behavioral/Framing prediction): Including a "deal-terms" slide (i.e., desired valuation, capital to be raised, equity to be sold) in a pitch deck will decrease investor favorability toward a startup.

Methods

We conducted a randomized controlled trial to examine the impact of including a single, experimentally manipulated "deal-terms" slide in the pitch decks of pre-seed-stage startups presented to investors. The sample consisted of 1,008 investor evaluations. We obtained these data by recruiting 88 investors (mean age of 28.1, 66 percent of whom were male) who had previously evaluated at least one early-stage startup. Following informed consent, each investor agreed to evaluate 12 pitch decks, for a total of 1,056 potential evaluations. Forty-five evaluations were not completed due to a software issue that prevented the display of the deck, resulting in 1,008 completed evaluations used in our analysis.

To ensure ecological validity, participants reviewed pitch decks from startups that had recently participated in the Techstars accelerator program. The 12 ventures varied substantially in industry and business model, providing heterogeneity and realism. We used the existing pitch decks developed by the entrepreneurs for the control group. For the experimental group, we did not modify any parts of the existing pitch decks; however, we added a single slide within the deck

that included quantitative information on deal terms, ensuring that it fully adhered to the deck's visual identity and mirrored the overall narrative style.

We adopted a multi-factor-level design. The first level was the addition (i.e., the experimental treatment) or non-addition (i.e., the control) of the deal slide. When the deal information was manipulated, we also created favorable (i.e., high) and unfavorable (i.e., low) versions of the deal from the investor's perspective. These versions varied across four key dimensions of the deal: the amount of money raised, the amount already committed, the venture's valuation, and the ownership percentage sold to investors. We used pitch decks from three different companies for each of these four dimensions. Because there were three possible versions of each pitch deck (i.e., control, high, and low), investors were randomly assigned to one of three conditions for each of the 12 decks with the following probabilities: 0.50 (control), 0.25 (high), and 0.25 (low). Each investor reviewed the pitch decks online, in random order, through the Qualtrics survey platform.

After viewing each pitch deck, investors completed a validated investment-preference scale (Shane et al., 2020) including 7-point Likert-scale items (1 = strongly disagree, 7 = strongly agree) such as: "*This venture constitutes a viable entrepreneurial opportunity,*" "*I would pursue a follow-up meeting to learn more about the venture,*" "*I would recommend this opportunity to a co-investor I have invested with in the past,*" and "*I would ask to see the venture's data room.*" Respondents were then asked to estimate the likelihood that the venture would achieve a successful exit at four valuation levels (USD 50 million, USD 100 million, USD 500 million, and USD 1 billion), using probability estimates ranging from 0 to 100 percent in 5-point increments. Finally, investors rated how good or bad they thought the pitch was on a 7-point scale (1 = extremely bad, 7 = extremely good).

We computed four dependent variables. *Success Odds* was calculated as the sum of the probabilities that the venture would achieve a successful exit at each of the four valuation levels (Cronbach's α = .92). *Good Business* was the mean of responses to "*This venture constitutes a viable entrepreneurial opportunity,*" "*This venture offers an attractive business model,*" and "*In your opinion, how good/bad was the pitch?*" (α = .86). *Next Steps* was the mean of responses to "*I would pursue a follow-up meeting to learn more about the venture,*" "*I would initiate due diligence on the venture,*" and "*I would ask to see the venture's data room*" (α = .92). *Share with Others* was the mean of responses to "*I would recommend this opportunity to a co-investor I have invested with in the past*" and "*I would recommend that my partner(s) take a look at the venture*" (α = .90).

The predictor variable was a dummy variable, coded 1 if the respondent was randomly assigned the manipulated deck (i.e., one containing the deal-terms slide) and 0 if assigned the control deck. To confirm that the effect was due to the presence of the additional slide rather than the favorability of the deal, we also estimated alternative models that distinguished between the high- and low-term conditions and the control.

Results

Ordinary least squares regressions with robust standard errors clustered by respondent show that the inclusion of the "deal-terms" slide had a significant negative effect on investor evaluations across all four dependent variables. Table 1 shows our main results for *Success Odds*.[1] In Model 1, we introduce the treatment. The random assignment of an additional slide on "the deal" has a negative effect ($\beta = -.11$, $t = 3.48$, $p < .001$) on the investors' assessment of the venture's success odds. In Model 2, we control for the deal term included. The random assignment

[1] For all regressions in this paper, we also ran a version that controlled for investor demographics to ensure that our randomization efforts were random. In the interest of space, we do not report those additional regressions but are available upon request. In no cases did the investor demographics alter the statistical significance or economic magnitude of the results of our randomized treatment.

of an additional slide on "the deal" again has a negative effect ($\beta = -.11$, $t = 3.48$, $p < .001$). In Model 3, we control for the pitch decks used. When investors are randomly assigned an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.11$, $t = 3.43$, $p < .001$). In sum, the random assignment of a deal term slide to a pitch deck reduces investors' assessment of the odds of the venture's success, providing support for H1b.

[Insert Table 1 About Here]

Table 2 shows our main results for *Next Steps*. In Model 1, we introduce the treatment. When we randomly assign an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.16$, $t = 4.99$, $p < .001$). In Model 2, we control for the deal terms used in the experiment. When we randomly assign an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.16$, $t = 5.09$, $p < .001$). In Model 3, we control for the pitch decks used. an additional slide is randomly assiged that manipulation has a negative effect ($\beta = -.11$, $t = 5.12$, $p < .001$). That is, the random assignment of a deal term slide to a pitch deck reduces the investors' willingness to initiate the next steps, providing support for H1b.

[Insert Table 2 About Here]

Table 3 shows our main results for *Good Business*. As above, in Model 1, we introduce the treatment: When investors are randomly assigned an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.12$, $t = 3.72$, $p < .001$). In Model 2, we control for the deal terms included. When investors are randomly assigned an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.12$, $t = 3.79$, $p < .001$). In Model 3, we control for the pitch decks used: When investors are randomly assigned an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.12$, $t = 3.80$, $p < .001$). We can therefore conclude that

the random assignment of a deal term slide to a pitch deck reduces the investors' assessment of the venture's quality, providing support for H1b.

[Insert Table 3 About Here]

Finally, Table 4 presents our main results for *Share with Others*. As shown in Model 1, when investors are randomly assigned an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.15$, $t = 4.68$, $p < .001$). In Model 2, we control for the deal terms used. When investors are randomly assigned an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.15$, $t = 3.76$, $p < .001$). As before, in Model 3, we control for the pitch decks used. When investors are randomly assigned an additional slide on "the deal," that manipulation has a negative effect ($\beta = -.15$, $t = 4.79$, $p < .001$). Collectively, these models indicate that the random assignment of a deal term slide to a pitch deck reduces the investors' willingness to share the investment opportunity, supporting H1b.

[Insert Table 4 About Here]

Next, we conducted several robustness checks to confirm the validity of our findings. First, we confirmed that the negative effect of the deal terms slide came from the added slide, and not from the desirability of the deal. We reran the regressions using favorable versions (i.e., high condition) and unfavorable versions (i.e., low condition) of the manipulation for investors.[2] As Table 5 shows, we found that random assignment of both favorable-to-investors and unfavorable-to-investors manipulations of the deal terms had negative, substantive, and statistically significant effects on the perception of the odds of a successful exit. We also found that random assignment of both favorable-to-investors and unfavorable-to-investors manipulations of the deal terms had negative, substantive, and statistically significant effects on the perception of the business concept

[2] In the interest of space, here, we summarize the findings of these tests rather than produce tables for all of them.

being good (Table 6). Similarly, we found that the random assignment of both deal term manipulations had negative, substantive, and statistically significant effects on willingness to take the next steps in the evaluation process (Table 7). Finally, random assignment of both favorable-to-investors and unfavorable-to-investors manipulations of the deal terms had negative, substantive, and statistically significant effects on the willingness to share the deal with other investors (Table 8). These effects were present in regression models without controls, with controls for different dimensions of deal terms, and with pitch deck controls. Altogether, we found robust evidence that the randomized addition of the deal term slide, not the desirability of those terms for investors, had an adverse effect on investors.

[Insert Tables 5 to 8 About Here]

Second, we confirmed that the specific dimensions of the deal terms manipulated did not influence the outcome. The negative effect of deal term inclusion was present regardless of whether the deal terms focused on valuation, ownership offered, the amount of money sought, or the amount of capital already committed. We found economically substantive and statistically significant effects for all four dimensions when the deal terms were focused on valuation and round size; for three of the four dependent variables (i.e., *Success Odds*, *Share with Others*, and *Good Business*) when the deal terms were focused on the amount committed; and, for two of the dependent variables (i.e., *Success Odds* and *Next Steps*) when the deal terms were focused on ownership stake.

Third, we ran the analysis at the item level to confirm that the effect of the manipulation on the construct was not artifactual. We found a negative effect of the randomized addition of a deal terms slide on investors' perception of the venture's success odds. This effect held in regressions controlling for both the pitch decks and deal factors. It was significant for three of the

four items used to test the hypothesis: the odds of a USD 50 million exit, a USD 100 million exit, and a USD 500 million exit, but not for the odds of a USD 1 billion exit.[3]

The negative effect of the randomized addition of a deal terms slide on investors' perception of the venture's quality, controlling for both the pitch decks and deal factors, was present for the scales' items "pitch quality", "viable opportunity", and "attractive model". The negative effect of the randomized addition of a deal terms slide on investors' perception of the next steps was in regressions controlling for both the pitch decks and deal factors for the scales' items "schedule a follow-up meeting", "initiate diligence", and "ask for access to a data room." The negative effect of the randomized addition of a deal terms slide on investors' willingness to share deals in regressions controlling for both pitch decks and deal factors is present for both "sharing with co-investors" and "sharing with partners". In sum, across all our analyses we found consistent support for H1b. These results indicate that investors rated startups less favorably when the pitch deck included a slide specifying the desired valuation, the amount of capital to be raised, and the equity to be sold. The direction and magnitude of the coefficients were consistent across model specifications, controlling for the deal-term dimension (i.e., ownership, committed capital, round size, valuation) and company fixed effects. The results remain substantively identical when estimated using random-effects models with investor and company clustering.

The effect was driven by the inclusion of quantitative deal information itself, not by whether the terms were favorable or unfavorable to the investor. Robustness checks confirmed

[3] We believe that the reason the USD 1 billion exit prediction did not hold was because of the very low likelihood of that outcome. According to data from CB Insights (https://www.cbinsights.com/research/unicorn-conversion-rate), among U.S. companies that raised their first early-stage financing round (angel/seed/Series A) in 2008-2014, less than 1% ended up becoming unicorns (valued ≥ USD1 billion) in later years, with later cohorts showing even lower rates (e.g., ~0.25% for 2012).

the stability of the results across alternative model specifications, exclusion of outliers, and inclusion of deck-order and investor-experience controls. The negative main effect persisted in every specification, and in analyses using investor-level controls (i.e., previous experience, prior investment count, and demographic variables) produced consistent results: The inclusion of these covariates did not substantively alter either the magnitude or the statistical significance of the treatment effect.

Study 2

Conceptual Framework and Hypotheses

Study 1 demonstrated that including a slide specifying quantitative deal terms (i.e., the desired valuation, capital to be raised, and equity to be sold) produced a systematic change in investor judgments. Study 2 investigates the underlying psychological mechanisms driving this effect. The behavioral decision-making literature provides several plausible theoretical explanations, each emphasizing a distinct psychological process through which numerical information can influence evaluation under uncertainty. Moreover, experimental evidence shows that even in seemingly competitive or structured environments, decision-makers mis-evaluate outcomes when noisy signals are present (e.g., Cooper & Fang, 2008), suggesting similar distortions may occur in early-stage investment evaluation. We consider and test seven alternative accounts as detailed below.

Ambiguity Aversion. Ambiguity aversion refers to the tendency to prefer options with known probabilities over those with unknown or imprecise likelihoods (Ellsberg, 1961; Camerer & Weber, 1992). This preference arises due to discomfort with missing or unreliable information, which can lead to the avoidance of ambiguous options even when the expected outcomes are statistically equal (Halevy, 2007; Trautmann & van de Kuilen, 2015). Research on decision-making under uncertainty shows that individuals' ambiguity attitudes and confidence biases

systematically shape strategic choices in uncertain environments (Gutierrez, Åstebro, & Obloj, 2020), reinforcing the idea that early-stage investors may respond strongly to informational cues that reduce perceived ambiguity.

In entrepreneurial finance and early-stage investment settings, where uncertainty is high, data is scarce, and track records are limited, ambiguity aversion becomes especially prominent. Investors evaluating nascent ventures frequently decide in the absence of clearly defined risks, which can trigger cognitive discomfort and lead to more conservative choices (Fox & Tversky, 1995). In such environments, quantitative deal information can play a crucial role in mitigating perceived ambiguity. By providing structured numerical representations of future outcomes, quantitative disclosures help reduce informational gaps and clarify the boundary conditions under which a startup might succeed or fail. Moreover, providing quantitative data can enhance investors' perceived competence and control, thereby reducing the psychological discomfort associated with ambiguity (Curley, Yates, & Abrams, 1986).

Psychological Reactance. Psychological reactance refers to the motivational resistance that arises when individuals perceive a threat to their freedom of choice or autonomy (Brehm, 1966). This theory suggests that when people feel their options are being restricted, they experience discomfort and are motivated to restore their sense of freedom, often by resisting or rejecting the imposed constraint (Miron & Brehm, 2006). In decision-making settings, even subtle cues that limit perceived agency can provoke reactance responses, including negative affect, reduced compliance, or a devaluation of the restricted option (Rains, 2013). In the context of early-stage investing, explicit deal terms may be interpreted by investors as constraining their ability to negotiate or tailor terms to their preferences. This perception could trigger psychological reactance, especially among experienced investors accustomed to negotiating deal structures with

substantial autonomy (Pavey & Sparks, 2009). The presence of fixed terms might reduce the perceived relational openness between entrepreneur and investor, framing the entrepreneur as controlling or overly prescriptive (Nagy, Pollack, Rutherford, & Lohrke, 2012).

Signaling Theory. Signaling theory addresses how individuals reduce information asymmetries in decision-making contexts by relying on observable cues to infer unobservable attributes (Spence, 1973; Connelly, Certo, Ireland, & Reutzel, 2011). In contexts where quality is not directly observable, such as early-stage venture investment, signals become critical mechanisms for assessing credibility, competence, and preparedness (Bergh, Connelly, Ketchen, & Shannon, 2014). Early-stage entrepreneurs, who may lack established track records, must strategically manage the signals they emit through pitch content, presentation style, and deal structuring choices. The inclusion of deal terms may serve as a deliberate signal of founder competence, strategic foresight, and readiness for partnership (Petty & Gruber, 2011). This specificity can suggest that the founder has a clear plan and understands investment norms, which could increase investor favorability (Janney & Folta, 2003).

Cognitive Load. Cognitive load theory suggests that people have limited working memory resources, and decision quality declines when the amount or complexity of information exceeds this capacity (Sweller, 1988). Cognitive load can be related to task complexity, how information is presented, and/or related to learning and schema construction. Motivation and self-regulatory capacity fluctuate in systematic ways that influence individuals' willingness to exert cognitive effort (Dai, Milkman, & Riis, 2015), leaving investors more reliant on simplified decision rules under conditions of strain or overload. Moreover, because incentive structures shape how decision-makers allocate attention and effort when evaluating alternatives (Larkin, 2014), investors may similarly rely on simplified heuristics when cognitive effort is costly. In high-stakes decision

environments such as early-stage investing, excessive cognitive load, often generated by complex or dense numerical content, may impair processing efficiency and undermine judgment quality (Sweller & Chandler, 1991).

Decision Avoidance. Decision avoidance refers to people's tendency to postpone, delegate, or altogether evade making choices, particularly when decisions are complex, emotionally charged, or consequential (Tversky & Shafir, 1992; Anderson, 2003). In such contexts, the psychological burden associated with making a choice can lead to inaction, even when a decision is clearly required. This behavior is not merely due to informational insufficiency, and is often instead a response to perceived responsibility, anticipated regret, or the psychological cost of making a potentially irreversible or high-stakes choice (Luce, 1998).

Anchoring and Adjustment. Anchoring and adjustment is a cognitive heuristic in which individuals rely heavily on initial numerical values—anchors—when making subsequent judgments, even if those anchors are arbitrary or irrelevant (Tversky & Kahneman, 1974). Once an anchor is set, adjustments made in light of additional information tend to be insufficient, leading to biased evaluations (Epley & Gilovich, 2006). In early-stage investing, quantitative deal information can serve as potent anchors. These initial values may influence how investors assess the overall attractiveness of the opportunity, potentially distorting judgments of fair value, expected return, or venture potential (Simon, Houghton, & Aquino, 2000).

Loss Aversion. Loss aversion, a key concept of prospect theory, explains how people tend to feel the pain of losses more strongly than the pleasure of equivalent gains (Kahneman & Tversky, 1979). In this framework, people are more sensitive to potential downside outcomes and tend to overweight them in decision-making. This asymmetry can distort judgments in contexts

involving risk and uncertainty, such as early-stage investing (Barberis, 2013). The inclusion of explicit deal terms may increase the salience of potential losses or unfavorable outcomes. Accordingly, we formulate a set of competing hypotheses:

H2a (Ambiguity aversion): Perceived ambiguity will mediate the relationship between deal-term inclusion and investor favorability.

H2b (Psychological reactance): Psychological reactance will mediate the relationship between deal-term inclusion and investor favorability.

H2c (Signaling): Perceived signaling strength will mediate the relationship between deal-term inclusion and investor favorability.

H2d (Cognitive load): Cognitive load will mediate the relationship between deal-term inclusion and investor favorability.

H2e (Decision avoidance): Decision avoidance will mediate the relationship between deal-term inclusion and investor favorability.

H2f (Anchoring): Anchoring effects will mediate the relationship between deal-term inclusion and investor favorability.

H2g (Loss aversion): Loss aversion will mediate the relationship between deal-term inclusion and investor favorability.

In this study, no a priori prediction is made as to which of the mechanisms will dominate. Accordingly, Study 2 empirically tests all seven potential mediators within a single multilevel mediation framework. Evidence of the most significant indirect effect of one variable over the others will identify the construct that most plausibly represents the psychological mechanism through which quantitative disclosure influences investor judgments.

Methods

Data were collected from a sample of 216 participants (Mage = 34.8 years, SD = 8.2; 47% female) recruited through Prolific. Prolific's built-in quality assurance tools were used to ensure participant authenticity and data reliability. As in Study 1, all participants were pre-screened to ensure they had previous exposure to entrepreneurial or investment decision-making. Each participant evaluated 12 independent startup pitches, yielding a total of 2,592 pitch-level observations, our unit of analysis. Observations were nested within participants, forming a two-level dataset with pitches at the first level and participants at the second level.

As in Study 1, we employed a randomized controlled design in which we manipulated the presentation of deal terms in the pitch materials. Each participant viewed versions of the same startup pitch that varied in the numerical values presented on the "deal-term" slide. The manipulation was fully within-subjects: Every participant evaluated both "high" and "low" deal-term conditions, coded as 1 and 0, respectively. High conditions displayed higher numerical values (e.g., higher valuations, larger round sizes, greater ownership stakes), whereas low conditions presented more conservative figures. Each participant viewed three pitches per deal-factor category—ownership, committed capital, round size, and valuation—resulting in twelve trials per participant. The order of presentation was randomized, to control for fatigue or learning effects.

The independent variable was the randomized numerical framing of the deal terms, dummy-coded (0 = low, 1 = high). High values were intended to represent more aggressive, entrepreneur-favorable deal positions, while low values reflected conservative or investor-favorable terms (Feld & Mendelson, 2016; Metrick & Yasuda, 2011). These values were embedded directly in the slides to maintain ecological validity and visual continuity with the original deck templates used in Study 1.

Our focal mediators were drawn from behavioral decision research, as discussed above. Perceived ambiguity was measured using four Likert-scale items assessing the degree to which the information in the pitch was ambiguous, unclear, or incomplete ($\alpha = .87$). Examples include: "*For this deck, ambiguous or missing key metrics made me uneasy*" and "*Not knowing exactly how certain figures were derived made me reluctant to trust them.*"

In addition to ambiguity, we measured psychological reactance, cognitive load, decision avoidance, signaling, anchoring, and loss aversion using brief validated scales derived from prior studies in organizational and behavioral decision-making research. All mediators were rated across four items on 7-point Likert scales, with the ratings aggregated at the pitch level for inferential analysis (see Appendix 1 for scale items and related references).

Four dependent variables were identical to those in Study 1: perceived success odds, good business, next steps, and share with others. Each variable captured distinct dimensions of investor judgment following exposure to a pitch. Ratings were treated as first-level outcomes in the multilevel analyses.

Three demographic covariates were included at the second level to control for potential heterogeneity in judgment tendencies: participant age (in years), gender (0 = male, 1 = female), and self-reported prior investment experience. These controls accounted for individual differences unrelated to the experimental manipulation. Moreover, the four deal-term dimensions (i.e., ownership, committed capital, round size, and valuation) were included as grouping variables to assess whether the observed patterns generalized across distinct types of numerical framing.

Data inspection revealed minimal missing data (<1% overall). Continuous variables were grand-mean centered before analysis. All assumptions of multilevel linear modeling (i.e., normality, linearity, and independence of residuals) were met. Multicollinearity diagnostics were

conducted for all mediators and covariates using Variance Inflation Factors (VIFs), which were computed separately by deal-factor condition. Across all models, VIF values ranged from 1.03 to 1.89 (M = 1.30, SD = 0.24), well below conservative thresholds (VIF < 3). The ownership condition exhibited the highest correlation among cognitive and decision-avoidance variables (max VIF = 1.89), which remained acceptable. These checks confirmed that the mediators were satisfactorily independent and could be included simultaneously in mediation models without concern for collinearity.

Because multiple pitches were nested within participants, all models were specified as two-level hierarchical linear models with random intercepts for participants. Each mediation model estimated whether the effect of high versus low deal-term framing on investor judgments operated indirectly through each mediator. Bootstrapped confidence intervals (1,000 resamples, clustered by participant) were computed for all indirect effects to assess mediation robustness.

Indirect effects were computed as the product of the *a* path (High vs. Low → Mediator) and the *b* path (Mediator → Outcome); direct effects were captured by the *c'* path (High vs. Low → Outcome, controlling for the mediator). Each dependent variable—success odds, good business, next steps, and share with others—was modeled separately, as were each of the four deal-factor contexts. This strategy enabled consistent comparison with Study 1 while assessing whether the mediating processes generalized across ownership, committed capital, round size, and valuation manipulations.

Results

Table 9 presents the descriptive statistics and correlations for all variables included in Study 2. The table reports means, standard deviations, and intercorrelations for the dependent measures, the seven potential mediators, and the experimental condition variable.

[Insert Table 9 About here]

Table 10 reports the results of the multilevel regressions, estimating the *a* paths (i.e., the effects of the high versus low deal-term manipulation on each mediator). As shown, the manipulation significantly reduced perceived ambiguity ($\beta = -0.27$, $t = -4.21$, $p < .001$), indicating that the high deal-term condition increased perceived clarity relative to the low condition. Importantly, none of the other mediators—psychological reactance, signaling, cognitive load, decision avoidance, anchoring, or loss aversion—were significantly affected by the manipulation ($p > .10$ for all).

[Insert Table 10 About here]

Table 11 summarizes the b paths (i.e., the effects of each mediator on investor favorability measures, controlling for the manipulation). Perceived ambiguity significantly predicted investor favorability across all four dependent variables: Success Odds ($\beta = -0.21$, $t = -3.86$, $p < .001$), Good Business ($\beta = -0.24$, $t = -4.11$, $p < .001$), Next Steps ($\beta = -0.19$, $t = -3.57$, $p < .001$), and Share with Others ($\beta = -0.18$, $t = -3.12$, $p < .01$). Higher perceived ambiguity was associated with lower favorability.

[Insert Table 11 About here]

To test mediation, we estimated indirect effects for all seven mechanisms using bootstrapped multilevel models with 1,000 resamples clustered by participant. Table 12 reports the results for the ambiguity-aversion model. The indirect effect of deal-term condition on investor favorability through perceived ambiguity was statistically significant for all dependent measures, with 95% bootstrapped confidence intervals excluding zero. For Success Odds, the indirect effect was −0.06, 95% CI [−0.10, −0.03]; for Good Business, −0.07, 95% CI [−0.12, −0.04]; for Next Steps, −0.05, 95% CI [−0.09, −0.03]; and for Share with Others, −0.04, 95% CI [−0.08, −0.02].

These results indicate that reduced perceived ambiguity mediated the relationship between numerical framing of deal terms and investor favorability.

[Insert Table 12 About here]

For the six other candidate mediators—psychological reactance, signaling, cognitive load, decision avoidance, anchoring, and loss aversion—none of the indirect effects were statistically significant. In all cases, the 95% bootstrapped confidence intervals for the indirect effects contained zero, and the corresponding *a* or *b* paths were non-significant.[4]

To confirm that ambiguity aversion uniquely accounted for the observed effect, we estimated a multiple mediator model that included all seven mediators simultaneously. Only perceived ambiguity remained significant ($p < .001$); all others remained non-significant. Finally, the direct effect of the deal-term condition on investor favorability, controlling for perceived ambiguity, was reduced to non-significance, confirming full mediation.

[Insert Table 13 About here]

Overall, the results of Study 2 support H2a only and fail to support H2b through H2g. The combined evidence from Studies 1 and 2 suggests that including quantitative deal terms influences investor favorability primarily by altering perceived informational ambiguity, rather than triggering psychological reactance, cognitive load, decision avoidance, or other psychological mechanisms.

General Discussion

Across two experiments, using randomized controlled trial designs, we find convergent evidence that including quantitative deal terms in startup pitch decks systematically reduces investor favorability, and that this effect operates through perceived ambiguity. Study 1

[4] Since these mechanisms showed no evidence of mediation, detailed regression tables are omitted for the sake of parsimony. Full statistical results for these models are available upon request.

demonstrated the phenomenon causally: Investors rated otherwise identical startups less positively when a slide specified the desired valuation, capital to be raised, and equity to be sold. Study 2 then identified the psychological mechanism; among seven plausible mediators drawn from behavioral decision theory, only perceived ambiguity accounted for this relationship. Quantitative disclosure reduced the sense of informational vagueness surrounding an early-stage venture, and this reduction in perceived ambiguity, in turn, increased investors' willingness to engage.

Our findings contribute to the broader literature on entrepreneurial pitching by providing an integrative, mechanism-based explanation of investor judgment (Kalvapalle et al., 2024). Rather than treating investor bias as an unexplained outcome or limiting focus to social categories such as gender and race, we identify perceived informational ambiguity as a generalizable psychological mechanism that influences investor judgment and favorability across contexts. In doing so, we help bridge the bifurcation between entrepreneur-centric and investor-centric views of pitching, as recently described by Kalvapalle et al. (2024), by integrating both perspectives through a communicative lens.

These results advance understanding of how information presentation shapes early-stage investment judgments. Practitioner advice remains divided on whether founders should "pitch the deal." Some experienced investors caution that listing specific terms "*anchors expectations and turns off potential backers*" (Y Combinator, 2020; Startup School, 2021), while others argue that "*serious investors need to know the ask up front*" (Techstars, 2022; Feld & Mendelson, 2016). Both intuitions are empirically grounded in opposing cognitive forces: the clarity provided by numbers versus the constraints those numbers impose. Our findings reconcile this debate by showing that what matters is not "the numbers themselves" but their psychological impact on

perceived ambiguity. In early-stage investing, where uncertainty is high and verifiable metrics are scarce, any factor that alters perceived clarity can meaningfully shift evaluation outcomes.

The results also extend behavioral decision theory into a domain characterized by extreme uncertainty and social complexity. Most prior work on ambiguity aversion has examined abstract lotteries or financial gambles (Ellsberg, 1961; Camerer & Weber, 1992), often through laboratory experiments. Our findings not only hold strongly in terms of ecological validity but also demonstrate that similar mechanisms apply to social and strategic financial judgments about entrepreneurial ventures. Investors are not simply reacting to numerical content as information; they are reacting to the degree of uncertainty those numbers resolve. When deal terms clarify the informational environment, investors respond more favorably. When such numbers are absent, investors perceive the opportunity as more ambiguous and discount it accordingly.

Importantly, the evidence argues against several alternative explanations. Psychological reactance, cognitive load, decision avoidance, anchoring, signaling, and loss aversion—all theoretically viable mediators—did not show reliable indirect effects. The lack of support for these mechanisms suggests that investors' reactions are not driven by resistance to constraint, information overload, or emotional loss sensitivity. Instead, the consistent indirect pathway through perceived ambiguity highlights the centrality of uncertainty perception in early-stage decision-making.

Together, the two studies highlight how seemingly minor presentation choices can significantly impact investor judgments by altering the decision context rather than the substantive content of the opportunity. In environments where information is sparse, disclosure structure acts as a form of sensemaking. Founders' choices about what to reveal and how to frame it implicitly guide investors' cognitive processing. The present results suggest that founders who clearly

articulate quantitative parameters—without overspecifying or exaggerating precision—help potential investors interpret uncertainty as structured rather than chaotic. Related work shows that when people operate under high external expectations, even small negative cues or early signals of difficulty can trigger disengagement or withdrawal due to embarrassment and impression-management concerns (Dai, Dietvorst, Tuckfield, Milkman, & Schweitzer, 2018), reinforcing the idea that investors may be unusually sensitive to informational signals in high-stakes evaluation contexts.

Several limitations qualify our conclusions. First, both studies use controlled experimental designs with professional or semi-professional investors evaluating standardized pitch materials. Real-world fundraising involves richer interpersonal and contextual cues that may moderate these effects. Future research should investigate whether perceived ambiguity has a similar effect in, for example, live pitch settings or investor–founder negotiations. Second, while the two-study design isolates perceived ambiguity as the primary mediator, other unmeasured constructs—such as trust formation or epistemic confidence—may covary with ambiguity perceptions and contribute to the observed pattern. Third, our manipulations focused on the presence and magnitude of quantitative deal terms; other disclosure elements (e.g., traction metrics, team qualifications) could potentially interact with numerical framing to shape judgments in complex ways.

Despite these limitations, our findings have rich implications for both theory and practice. Theoretically, they highlight perceived ambiguity as a central psychological construct in entrepreneurial finance, bridging the literatures on behavioral decision-making and venture evaluation. Practically, they suggest that founders' disclosure strategies should be guided by how potential investors interpret uncertainty, not simply by the amount of information shared. A

concise and transparent presentation of valuation and ownership expectations can improve investor receptivity by clarifying uncertainty rather than amplifying it.

In conclusion, the combined evidence from Studies 1 and 2 reveals that quantitative "deal-term" disclosure exerts a measurable and psychologically interpretable effect on investor evaluations. The decision to "pitch the deal" is not merely a matter of stylistic preference. It is a strategic lever that influences how investors perceive and construe uncertainty. By clarifying, rather than concealing ambiguity, founders can frame their opportunities in a way that elicits more favorable and confident investor responses.

References

Anderson, C. J. (2003). The psychology of doing nothing: Forms of decision avoidance result from reason and emotion. *Psychological Bulletin, 129*(1), 139–167. https://doi.org/10.1037/0033-2909.129.1.139

Barberis, N. C. (2013). Thirty years of prospect theory in economics: A review and assessment. *Journal of Economic Perspectives*, 27(1), 173-196.

Bergh, D. D., Connelly, B. L., Ketchen Jr, D. J., & Shannon, L. M. (2014). Signalling theory and equilibrium in strategic management research: An assessment and a research agenda. *Journal of Management Studies*, 51(8), 1334-1360.

Bernstein, S., Korteweg, A., & Laws, K. (2017). Attracting early-stage investors: Evidence from a randomized field experiment. *Journal of Finance, 72*(2), 509–538. https://doi.org/10.1111/jofi.12469

Brehm, J. W. (1966). *A theory of psychological reactance.* New York, NY: Academic Press.

Brooks, A. W., Huang, L., Kearney, S. W., & Murray, F. E. (2014). Investors prefer entrepreneurial ventures pitched by attractive men. *Proceedings of the National Academy of Sciences, 111*(12), 4427–4431. https://doi.org/10.1073/pnas.1321202111

Camerer, C. F., & Weber, M. (1992). Recent developments in modeling preferences: Uncertainty and ambiguity. *Journal of Risk and Uncertainty, 5*(4), 325–370. https://doi.org/10.1007/BF00122575

Connelly, B. L., Certo, S. T., Ireland, R. D., & Reutzel, C. R. (2011). Signaling theory: A review and assessment. *Journal of Management, 37*(1), 39–67. https://doi.org/10.1177/0149206310388419

Cooper, D. J., & Fang, H. (2008). Understanding overbidding in second price auctions: An experimental study. *The Economic Journal*, *118*(532), 1572-1595.

Curley, S. P., Yates, J. F., & Abrams, R. A. (1986). Psychological sources of ambiguity avoidance. *Organizational Behavior and Human Decision Processes*, 38(2), 230-256.

Dai, H., Dietvorst, B. J., Tuckfield, B., Milkman, K. L., & Schweitzer, M. E. (2018). Quitting when the going gets tough: A downside of high performance expectations. *Academy of Management Journal*, *61*(5), 1667-1691.

Dai, H., Milkman, K. L., & Riis, J. (2014). The fresh start effect: Temporal landmarks motivate aspirational behavior. *Management Science*, *60*(10), 2563-2582.

Dai, H., Milkman, K. L., & Riis, J. (2015). Put your imperfections behind you: Temporal landmarks spur goal initiation when they signal new beginnings. *Psychological Science*, *26*(12), 1927-1936.

Ellsberg, D. (1961). Risk, ambiguity, and the Savage axioms. *Quarterly Journal of Economics, 75*(4), 643–669. https://doi.org/10.2307/1884324

Epley, N., & Gilovich, T. (2006). The anchoring-and-adjustment heuristic: Why the adjustments are insufficient. *Psychological Science*, 17(4), 311-318.

Feld, B., & Mendelson, J. (2016). *Venture deals: Be smarter than your lawyer and venture capitalist* (3rd ed.). Hoboken, NJ: Wiley.

Fox, C. R., & Tversky, A. (1995). Ambiguity aversion and comparative ignorance. *The Quarterly Journal of Economics,* 110(3), 585-603.

Gutierrez, C., Åstebro, T., & Obloj, T. (2020). The impact of overconfidence and ambiguity attitude on market entry. *Organization Science*, 31(2), 308-329.

Halevy, Y. (2007). Ellsberg revisited: An experimental study. *Econometrica*, 75(2), 503-536.

Huang, L., & Pearce, J. L. (2015). Managing the unknowable: The effectiveness of early-stage investor gut feel in entrepreneurial investment decisions. *Administrative Science Quarterly, 60*(4), 634–670. https://doi.org/10.1177/0001839215597270

Janney, J. J., & Folta, T. B. (2003). Signaling through private equity placements and its impact on the valuation of biotechnology firms. *Journal of Business Venturing*, 18(3), 361-380.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica, 47*(2), 263–291. https://doi.org/10.2307/1914185

Kalvapalle, V., Phillips, N., & Cornelissen, J. (2024). Entrepreneurial pitching: Toward a theoretical synthesis of a fragmented literature. *Academy of Management Annals, 18*(1), 177–212. https://doi.org/10.5465/annals.2022.0066

Kanze, D., Huang, L., Conley, M. A., & Higgins, E. T. (2018). We ask men to win and women not to lose: Closing the gender gap in startup funding. *Academy of Management Journal, 61*(2), 586–614. https://doi.org/10.5465/amj.2016.1215

Larkin, I. (2014). The cost of high-powered incentives: Employee gaming in enterprise software sales. *Journal of Labor Economics*, *32*(2), 199-227.

Lichtenstein, S., & Slovic, P. (2006). *The construction of preference.* Cambridge University Press.

Luce, M. F. (1998). Choosing to avoid: Coping with negatively emotion-laden consumer decisions. *Journal of Consumer Research*, 24(4), 409-433.

Metrick, A., & Yasuda, A. (2011). Venture capital and the finance of innovation (2nd ed.). Hoboken, NJ: Wiley.

Miron, A. M., & Brehm, J. W. (2006). Reactance theory-40 years later. *Zeitschrift für Sozialpsychologie*, 37(1), 9-18.

Nagy, B. G., Pollack, J. M., Rutherford, M. W., & Lohrke, F. T. (2012). The influence of entrepreneurs' credentials and impression management behaviors on perceptions of new venture legitimacy. *Entrepreneurship Theory and Practice*, 36(5), 941-965.

Packard, M. D., Clark, B. B., & Klein, P. G. (2017). Uncertainty types and transitions in the entrepreneurial process. *Organization Science, 28*(5), 840–856. https://doi.org/10.1287/orsc.2017.1143

Pavey, L., & Sparks, P. (2009). Reactance, autonomy and paths to persuasion: Examining perceptions of threats to freedom and informational value. *Motivation and Emotion*, 33(3), 277-290.

Petty, J. S., & Gruber, M. (2011). In pursuit of the real deal: A longitudinal study of VC decision making. *Journal of Business Venturing, 26*(2), 172–188. https://doi.org/10.1016/j.jbusvent.2009.07.002

Rains, S. A. (2013). The nature of psychological reactance revisited: A meta-analytic review. *Human Communication Research*, 39(1), 47-73.

Shane, S., & Stuart, T. E. (2002). Organizational endowments and the performance of university start-ups. *Management Science, 48*(1), 154–170. https://doi.org/10.1287/mnsc.48.1.154.14280

Simon, M., Houghton, S. M., & Aquino, K. (2000). Cognitive biases, risk perception, and venture formation: How individuals decide to start companies. *Journal of Business Venturing*, 15(2), 113-134.

Slovic, P., Fischhoff, B., & Lichtenstein, S. (1977). Behavioral decision theory. *Annual Review of Psychology, 28*, 1–39. https://doi.org/10.1146/annurev.ps.28.020177.000245

Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics, 87*(3), 355–374. https://doi.org/10.2307/1882010

Startup School by Y Combinator. (2021). *Pitching investors: Dos and don'ts.* https://www.startupschool.org/videos

Steindl, C., Jonas, E., Sittenthaler, S., Traut-Mattausch, E., & Greenberg, J. (2015). Understanding psychological reactance: New developments and findings. *Zeitschrift für Psychologie, 223*(4), 205–214. https://doi.org/10.1027/2151-2604/a000222

Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science, 12*(2), 257–285. https://doi.org/10.1016/0364-0213(88)90023-7

Sweller, J., & Chandler, P. (1991). Evidence for cognitive load theory. *Cognition and Instruction*, 8(4), 351-362.

Techstars. (2022, April 6). *How to build your startup pitch deck.* Retrieved from https://www.techstars.com/newsroom/how-to-build-your-startup-pitch-deck

Trautmann, S. T., & Van De Kuilen, G. (2015). Ambiguity attitudes. *The Wiley Blackwell Handbook Of Judgment And Decision Making*, 2, 89-116.

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science, 185*(4157), 1124–1131. https://doi.org/10.1126/science.185.4157.1124

Tversky, A., & Shafir, E. (1992). Choice under conflict: The dynamics of deferred decision. *Psychological Science, 3*(6), 358–361. https://doi.org/10.1111/j.1467-9280.1992.tb00047.x

Wiltbank, R., Dew, N., Read, S., & Sarasvathy, S. D. (2009). Prediction and control under uncertainty: Outcomes in angel investing. *Journal of Business Venturing, 24*(2), 116–133. https://doi.org/10.1016/j.jbusvent.2007.11.004

Y Combinator. (2020, July 15). *Common pitch deck mistakes.* Retrieved from https://www.ycombinator.com/library/4A-common-pitch-deck-mistakes

Younkin, P., & Kuppuswamy, V. (2017). The colorblind crowd? Founder race and performance in crowdfunding. *Management Science, 64*(7), 3269–3287. https://doi.org/10.1287/mnsc.2017.2753.

Tables

Table 1. Effect of the Deal Term Treatment on Investors' Perceptions of the Venture's Odds of Success.

	Model 1: No Controls		**Model 2: Deal Term Controls**		**Model 3: Company Controls**	
Deal Term	-.11	***	-.11	***	-.11	***
Committed			-.08	*		
Ownership			-.02			
Valuation			-.05			
Company B					.01	
Company C					-.01	
Company D					-.02	
Company E					.01	
Company F					-.04	
Company G					.00	
Company H					.08	t
Company I					.14	***
Company J					.03	
Company K					-.05	
Company L					.06	
R-Squared	0.01		0.02		0.04	
F-Value	12.14	***	4.48	**	4.06	**
N	1,008		1,008		1,008	

Key: t = p < .10, * = p < .05, ** = p < .01, *** = p < .001.

Table 2. Effect of the Deal Term Treatment on Investors' Willingness to Take Next Steps.

	Model 1: No Controls		**Model 2: Deal Term Controls**		**Model 3: Company Controls**	
Deal Term	-0.16	***	-0.16	***	-0.11	***
Committed			-0.03			
Ownership			0.05			
Valuation			-0.07	t		
Company B					0.02	
Company C					0.02	
Company D					0.00	
Company E					0.00	
Company F					-0.08	*
Company G					0.08	
Company H					0.09	*
Company I					0.12	**
Company J					0.10	*
Company K					-0.02	
Company L					0.08	t
R-Squared	0.02		0.03		0.06	
F-Value	24.86	***	8.49	***	5.90	***
N	1,008		1,008		1,008	

Key: t = p < .10, * = p < .05, ** = p < .01, *** = p < .001.

Table 3. Effect of the Deal Term Treatment on Investors' Perception of Good Business.

	Model 1: No Controls		**Model 2: Deal Term Controls**		**Model 3: Company Controls**	
Deal Term	-.12	***	-.12	***	-.12	***
Committed			-.04			
Ownership			.02			
Valuation			-.07	t		
Company B					-.01	
Company C					-.02	
Company D					-.02	
Company E					.01	
Company F					-.08	t
Company G					.08	t
Company H					.13	**
Company I					.15	***
Company J					.05	
Company K					.01	
Company L					.06	
R-Squared	0.01		0.02		0.07	
F-Value	13.85	***	5.02	***	5.89	***
N	1,008		1,008		1,008	

Key: t = p <. 10, * = p < .05, ** = p < .01, *** = p < .001.

Table 4. Effect of the Deal Term Treatment on Investors' Willingness to Share the Opportunity.

	Model 1: No Controls		**Model 2: Deal Term Controls**		**Model 3: Company Controls**	
Deal Term	-0.15	***	-0.15	***	-0.15	***
Committed			-0.02			
Ownership			0.04			
Valuation			-0.05			
Company B					0.02	
Company C					0.03	
Company D					0.01	
Company E					0.02	
Company F					-0.06	
Company G					0.11	**
Company H					0.12	**
Company I					0.14	***
Company J					0.09	*
Company K					-0.01	
Company L					0.11	*
R-Squared	0.02		0.02		0.06	
F-Value	21.92	***	6.87	***	5.89	***
N	1,008		1,008		1,008	

Key: * = p < .05, ** = p < .01, *** = p < .001.

Table 5. Effect of High and Low Deal Term Treatment on Investors' Perceptions of the Venture's Odds of Success.

	Model 1: No Controls		**Model 2: Deal Term Controls**		**Model 3: Company Controls**	
High Deal	-.08	***	-.08	***	-.08	***
Low Deal	-.12	***	-.12	***	-.12	***
Committed			-.08	*		
Ownership			-.01			
Valuation			-.05			
Company B					.01	
Company C					-.01	
Company D					-.02	
Company E					.01	
Company F					-.04	
Company G					.01	
Company H					.08	t
Company I					.14	***
Company J					.03	
Company K					-.05	
Company L					.06	
R-Squared	0.01		0.01		0.05	
F-Value	6.65	***	3.78	**	3.83	**
N	1,008		1,008		1,008	

Key: t = p < .10, * = p < .05, ** = p < .01, *** = p < .001.

Table 6. Effect of High and Low Deal Term Treatment on Investors' Willingness to Take Next Steps.

	Model 1: No Controls		**Model 2: Deal Term Controls**		**Model 3: Company Controls**	
High Deal	-.16	***	-.16	***	-.16	***
Low Deal	-.13	***	-.13	***	-.13	***
Committed			-.03	*		
Ownership			.05			
Valuation			-.06	t		
Company B					.01	
Company C					-.02	
Company D					.00	
Company E					.00	
Company F					-.08	*
Company G					.08	t
Company H					.09	*
Company I					.12	**
Company J					.10	*
Company K					-.02	
Company L					.08	t
R-Squared	0.02		0.03		0.06	
F-Value	12.78	***	6.92	***	5.49	***
N	1,008		1,008		1,008	

Key: t = p < .10, * = p < .05, ** = p < .01, *** = p < .001.

Table 7. Effect of High and Low Deal Term Treatment on Investors' Perception of Good Business.

	Model 1: No Controls		**Model 2: Deal Term Controls**		**Model 3: Company Controls**	
High Deal	-.12	***	-.12	***	-.12	***
Low Deal	-.09	**	-.09	**	-.09	**
Committed			-.04			
Ownership			.02			
Valuation			-.07	t		
Company B					-.01	
Company C					-.02	
Company D					-.02	
Company E					.01	
Company F					-.08	t
Company G					.08	t
Company H					.13	***
Company I					.15	***
Company J					.05	
Company K					-.01	
Company L					.06	
R-Squared	0.01		0.02		0.06	
F-Value	7.25	***	4.14	**	5.49	***
N	1,008		1,008		1,008	

Key: t = p < .10, * = p < .05, ** = p < .01, *** = p < .001.

Table 8. Effect of High and Low Deal Term Treatment on Investors' Willingness to Share the Opportunity.

	Model 1: No Controls		**Model 2: Deal Term Controls**		**Model 3: Company Controls**	
High Deal	-.15	***	-.15	***	-.15	***
Low Deal	-.12	***	-.13	***	-.12	***
Committed			-.02			
Ownership			.04			
Valuation			-.04			
Company B					.02	
Company C					.03	
Company D					.02	
Company E					.02	
Company F					-.06	
Company G					.11	**
Company H					.12	**
Company I					.14	***
Company J					.09	*
Company K					-.01	
Company L					.11	*
R-Squared	0.02		0.02		0.05	
F-Value	11.14	***	5.55	***	5.46	***
N	1,008		1,008		1,008	

Key: t = p < .10, * = p < .05, ** = p < .01, *** = p < .001.

Table 9. Descriptive Statistics and Correlations (Study 2).

Variable	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12
1 Deal-term condition (1 = High)	0.50	0.50	–											
2 Perceived ambiguity	3.96	1.08	−.27***	–										
3 Psychological reactance	2.84	1.21	.04	−.06	–									
4 Signaling	4.91	1.07	.02	−.03	.07	–								
5 Cognitive load	3.12	1.02	.05	.09	.11	.04	–							
6 Decision avoidance	3.43	1.09	.03	.10	.12	.06	.22***	–						
7 Anchoring	3.97	1.14	.01	−.04	.03	.09	.07	.05	–					
8 Loss aversion	3.64	1.13	.02	.05	.08	.10	.06	.04	.03	–				
9 Success odds	1.72	0.84	−.09***	−.21***	−.02	.04	−.05	−.03	.00	−.01	–			
10 Good business	5.01	1.08	−.11***	−.24***	−.03	.05	−.06	−.04	.02	−.02	.73***	–		
11 Next steps	3.42	1.30	−.14***	−.19***	−.04	.06	−.05	−.03	.03	−.01	.68***	.65***	–	
12 Share with others	3.69	1.45	−.13***	−.18***	−.02	.07	−.04	−.02	.04	−.01	.66***	.61***	.74***	–

Key: * = p < .05, ** = p < .01, *** = p < .001. N = 2,592 pitch-level observations; 216 participants.

Table 10. Effects of Deal-Term Framing on Mediators (*a* paths; multilevel models with random intercepts for participants).

Mediator	**β**	**SE**	**t**	**p**
Perceived ambiguity	−0.27	0.06	−4.21	< .001
Psychological reactance	0.04	0.07	0.61	.54
Signaling	0.02	0.05	0.48	.63
Cognitive load	0.05	0.06	0.78	.44
Decision avoidance	0.03	0.06	0.55	.58
Anchoring	0.01	0.05	0.21	.84
Loss aversion	0.02	0.06	0.33	.74

Random intercept variance = 0.42; Residual variance = 0.71.

Table 11. Effects of Mediators on Investor Favorability Measures Controlling for Deal-Term Framing (*b* paths).

Mediator	**Success Odds β (t)**	**Good Business β (t)**	**Next Steps β (t)**	**Share with Others β (t)**
Perceived ambiguity	−0.21 (−3.86)***	−0.24 (−4.11)***	−0.19 (−3.57)***	−0.18 (−3.12)**
Psychological reactance	−0.04 (−0.88)	−0.03 (−0.71)	−0.05 (−1.02)	−0.02 (−0.55)
Signaling	0.05 (1.24)	0.07 (1.36)	0.04 (1.05)	0.05 (1.11)
Cognitive load	−0.02 (−0.54)	−0.03 (−0.61)	−0.02 (−0.49)	−0.01 (−0.33)
Decision avoidance	−0.05 (−0.91)	−0.04 (−0.82)	−0.03 (−0.77)	−0.02 (−0.61)
Anchoring	0.03 (0.69)	0.02 (0.58)	0.02 (0.63)	0.01 (0.45)
Loss aversion	−0.02 (−0.47)	−0.03 (−0.66)	−0.01 (−0.39)	−0.01 (−0.34)

Key: ** = p < .01, *** = p < .001.

Table 12. Indirect Effects of Deal-Term Condition on Investor Favorability Through Perceived Ambiguity.

Dependent Variable	**Indirect Effect ($a \times b$)**	**SE**	**95 % CI Lower**	**95 % CI Upper**
Success odds	−0.06	0.02	−0.10	−0.03
Good business	−0.07	0.02	−0.12	−0.04
Next steps	−0.05	0.02	−0.09	−0.03
Share with others	−0.04	0.02	−0.08	−0.02

Key: All indirect effects are significant, $p < .01$. Bootstrapped 95 % Cis; 1,000 resamples clustered by participant.

Table 13. Multiple-Mediator Model Including All Seven Mechanisms Simultaneously.

Mediator	**Indirect Effect (*a* × *b*)**	**SE**	**95 % CI Lower**	**95 % CI Upper**	**Sig.**
Perceived ambiguity	−0.06	0.02	−0.10	−0.03	**
Psychological reactance	−0.00	0.01	−0.03	0.02	n.s.
Signaling	0.01	0.01	−0.02	0.03	n.s.
Cognitive load	−0.01	0.01	−0.04	0.02	n.s.
Decision avoidance	−0.01	0.01	−0.04	0.02	n.s.
Anchoring	0.00	0.01	−0.02	0.02	n.s.
Loss aversion	−0.01	0.01	−0.03	0.02	n.s.

Key: n.s. = non significant; ** = p <.01.

Appendix

Table S1. Scales and Items Used for Mediators in Study 2.

Construct	Scale's Items	References
Ambiguity aversion	For this deck, ambiguous or missing key metrics made me uneasy. For this deck, I felt less confident when its key metrics weren't fully spelled out. Ambiguous information about this deck reduced my willingness to invest. Not knowing exactly how certain figures were derived made me reluctant to trust them.	Dimmock, S., Kouwenberg, R., Mitchell, O., & Peijnenburg, K. (2015). Ambiguity aversion and household portfolio choice. *Journal of Financial Economics*. Fox, C. R., & Tversky, A. (1995). Ambiguity aversion and comparative ignorance. *Quarterly Journal of Economics*, 110(3), 585–603. McLain, D. L. (1993). The MSTAT-I: A new measure of an individual's tolerance for ambiguity. *Educational and Psychological Measurement*, 53(1), 183-189.
Psychological reactance	The valuation request felt like pressure on my freedom to decide. I felt resistant to the deck's conclusions because they seemed overly controlling or one-sided.	Dillard, J. P., & Shen, L. (2005). On the nature of reactance and its role in persuasive health communication. *Communication Monographs*, 72(2), 144–168.

	For this deck, seeing those preset key metrics made me feel my freedom to decide was restricted. For this deck, I wanted to push back because I felt forced by its key metrics.	Hong, S. M. (1992). Hong's Psychological Reactance Scale: A further factor analytic validation. *Psychological Reports*, 70(2), 512-514. Hong, S. M., & Faedda, S. (1996). Refinement of the Hong psychological reactance scale. *Educational and Psychological Measurement*, 56(1), 173-182.
Signaling	For this deck, showing its key metrics increased my trust in the founders. For this deck, transparent key metrics made me view the pitch as more professional. For this deck, I felt more confident because its key metrics seemed honest. The clarity of the pitch's metrics made me believe the startup had nothing to hide.	Busenitz, L. W., Fiet, J. O., & Moesel, D. D. (2005). Signaling in venture capitalist–new venture team funding decisions: Does it indicate long-term venture outcomes? *Entrepreneurship Theory and Practice*, 29(1), 1–12. Spence, M.A. (1973), Job market signalling, *Quarterly Journal of Economics*, 87, 355–74. Spence, M.A. (2002), Signalling in retrospect and the informational structure of markets, *American Economic Review*, 92 (3), 434–59.

Cognitive load	For this deck, I did not have to work hard to understand those key metrics. (Reverse scored) Some parts of the pitch felt convincing without needing further analysis. I relied on my initial impressions rather than deeply evaluating all the deck's details. I accepted the deck's information quickly without scrutinizing alternative interpretations.	

Chaiken, S. (1980). Heuristic versus systematic information processing and the use of source versus message cues in persuasion. *Journal of Personality and Social Psychology*, 39(5), 752–766.

Petty, R. E., & Cacioppo, J. T. (1986). The elaboration likelihood model of persuasion. In *Advances in Experimental Social Psychology* (Vol. 19, pp. 123-205). Academic Press.

Reber, R., Schwarz, N., & Winkielman, P. (2004). Processing fluency and aesthetic pleasure: Is beauty in the perceiver's processing experience? *Personality and Social Psychology Review*, 8(4), 364-382.

Stanovich, K. E., & West, R. F. (2000). Individual differences in reasoning: Implications for the rationality debate? *Behavioral and Brain Sciences*, 23(5), 645–665.

Decision avoidance	For this deck, missing or vague key metrics made me want to delay my evaluation. For this deck, I hesitated to form a judgment because its key metrics were unclear. The pitch made me lean toward 'wait and see' rather than make a firm call. For this deck, missing or vague key metrics made me lean toward 'wait and see' rather than make a firm call.	Anderson, C. J. (2003). The psychology of doing nothing: Forms of decision avoidance result from reason and emotion. *Psychological Bulletin*, 129(1), 139–167. Janis, I. L., & Mann, L. (1977). Decision making: A psychological analysis of conflict, choice, and commitment. *Free Press.* Tversky, A., & Shafir, E. (1992). Choice under conflict: The dynamics of deferred decision. *Psychological Science*, 3(6), 358-361.
Anchoring and adjustment heuristic	For this deck, to what extent did your final judgment stay close to those key metrics? While evaluating this deck, how difficult was it for you to adjust your opinion away from those key metrics? I found myself referring back to the initial figures presented, even when considering unrelated aspects of the business. The initial numerical information shaped how I interpreted the rest of the pitch content.	Chapman, G. B., & Johnson, E. J. (1999). Anchoring, activation, and construction of values. *Organizational Behavior and Human Decision Processes*, 79(2), 115–153. Epley, N., & Gilovich, T. (2001). Putting adjustment back in the anchoring and adjustment heuristic: Differential processing of self-generated and experimenter-provided anchors. *Psychological Science*, 12(5), 391-396.

		Furnham, A., & Boo, H. C. (2011). A literature review of the anchoring effect. *The Journal of Socio-economics*, 40(1), 35-42. Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157), 1124–1131.
Loss aversion	Possible financial setbacks from investing in this startup felt harder to justify than missing out on its success. I focused more on what I could lose than on what I could gain by investing. Avoiding losses matters more to me than achieving gains in this deal. The risk of losing money on this deal felt more pressing than the possibility of a high return.	Benartzi, S., & Thaler, R. H. (1995). Myopic loss aversion and the equity premium puzzle. *Quarterly Journal of Economics*, 110(1), 73–92. Cabedo-Peris, J., Merino-Soto, C., Chans, G. M., & Martí-Vilar, M. (2024). Exploring the Loss Aversion Scale's psychometric properties in Spain. *Scientific Reports*, *14*(1), 15756. Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.

Note: Cronbach's alpha coefficients indicated good internal consistency for the mediator scales: anchoring, $\alpha = .84$; reactance, $\alpha = .79$; cognitive, $\alpha = .82$; ambiguity, $\alpha = .87$; decision avoidance, $\alpha = .74$; signaling, $\alpha = .81$.